

Locally global simulations of the convective overstability

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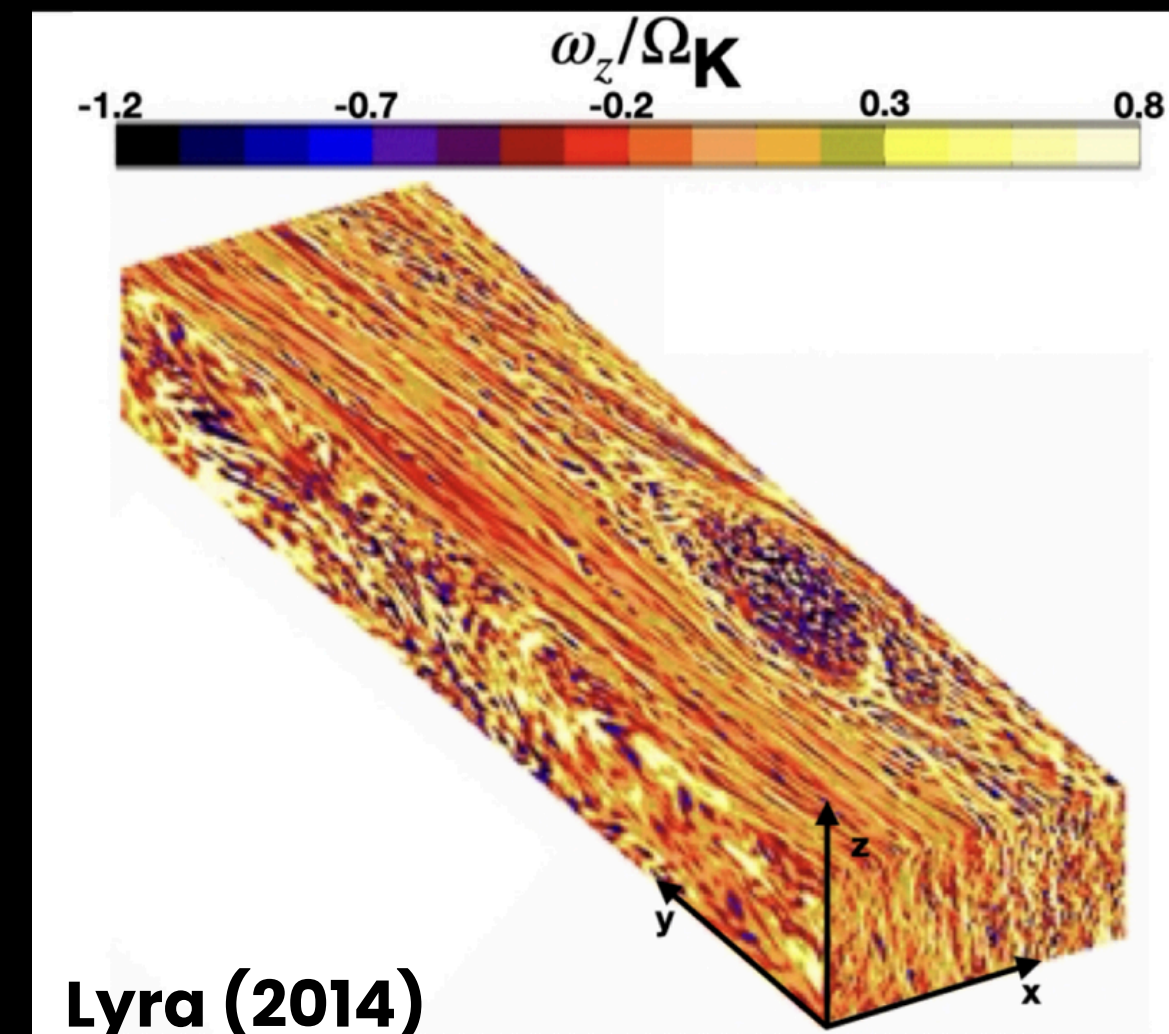
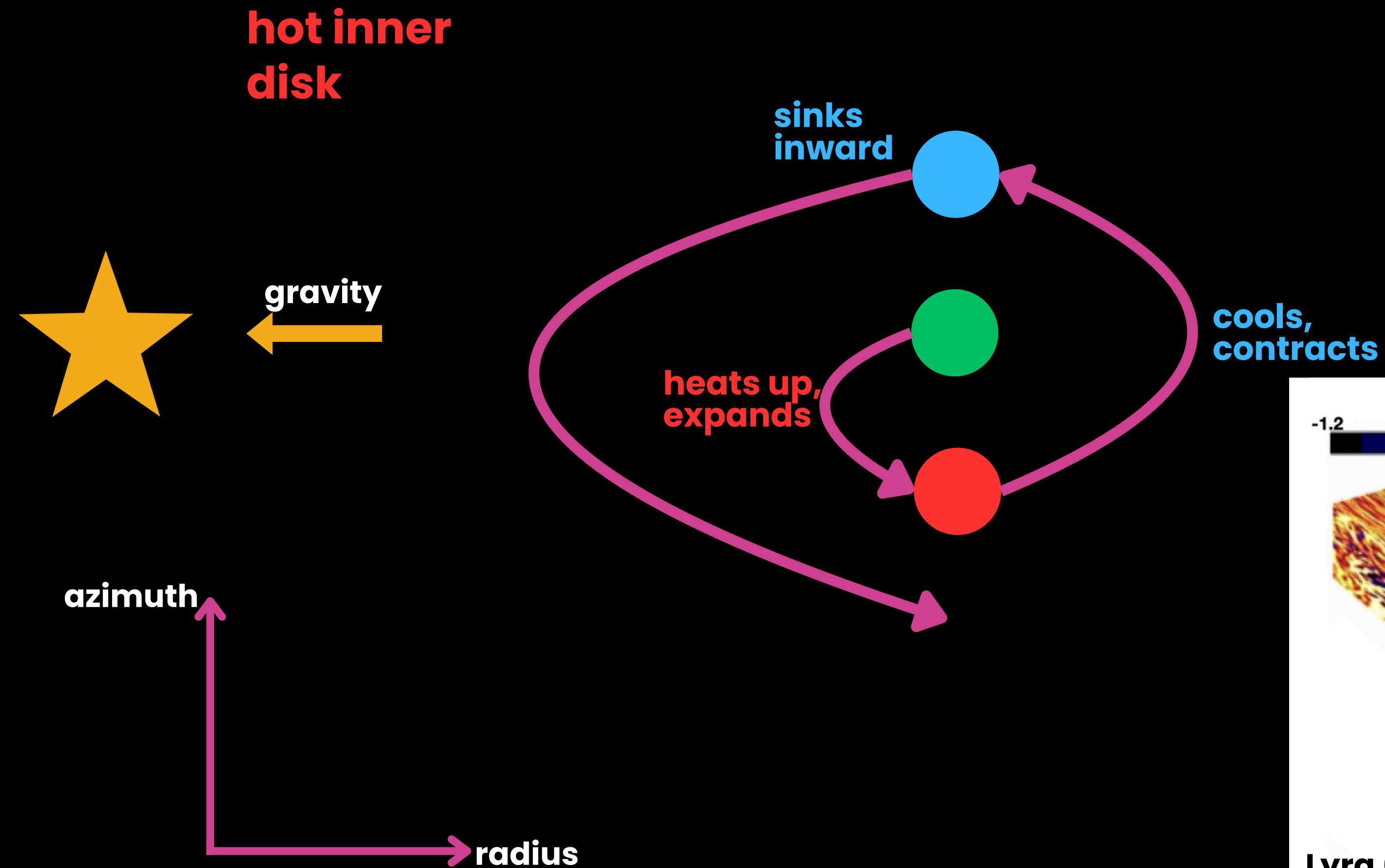


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Convective overstability (cos)



Midplane* instability criteria

- **Adverse radial buoyancy:**

$$N^2 \equiv -\frac{1}{\gamma\rho} \frac{\partial P}{\partial r} \frac{\partial S}{\partial r} < 0$$

→ usually need a negative entropy gradient, $dS/dr < 0$

- **Dynamical thermal timescales:**

$$t_{\text{cool}}\Omega \sim 1$$

*relaxed off midplane (Klahr 2026)

Local Boussinesq models

(Latter & Papaloizou, 2017)

slow dynamics

$$\nabla \cdot \mathbf{v}' = 0,$$

$$\frac{\partial \mathbf{v}'}{\partial t} + \mathbf{v}' \cdot \nabla \mathbf{v}' - \frac{3}{2} \Omega x \frac{\partial \mathbf{v}'}{\partial y} = 2\Omega v'_y \hat{x} - \frac{1}{2} \Omega v'_x \hat{y} - \nabla W - N^2 \theta \hat{x} + \nu \nabla^2 \mathbf{v}',$$

temperature evol.

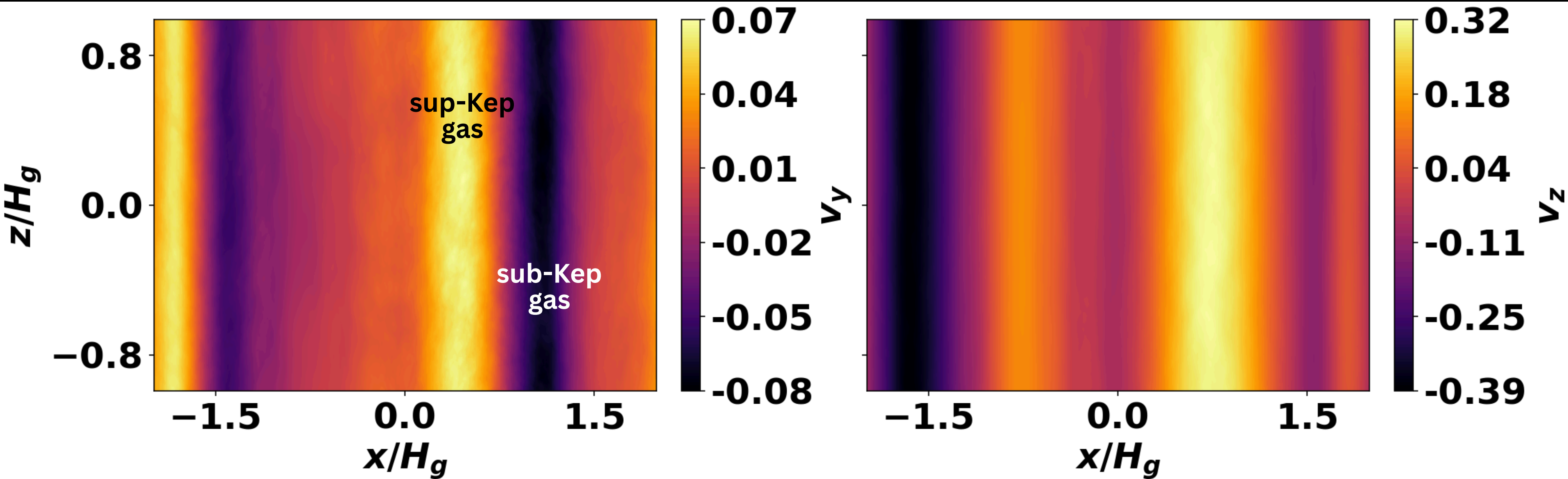
$$\frac{\partial \theta}{\partial t} + \mathbf{v}' \cdot \nabla \theta - \frac{3}{2} \Omega x \frac{\partial \theta}{\partial y} = v'_x - \xi \nabla^2 \theta,$$

buoyant forcing (input)

constant, thermal diffusion

see Lyra et al (2011) for a compressible version

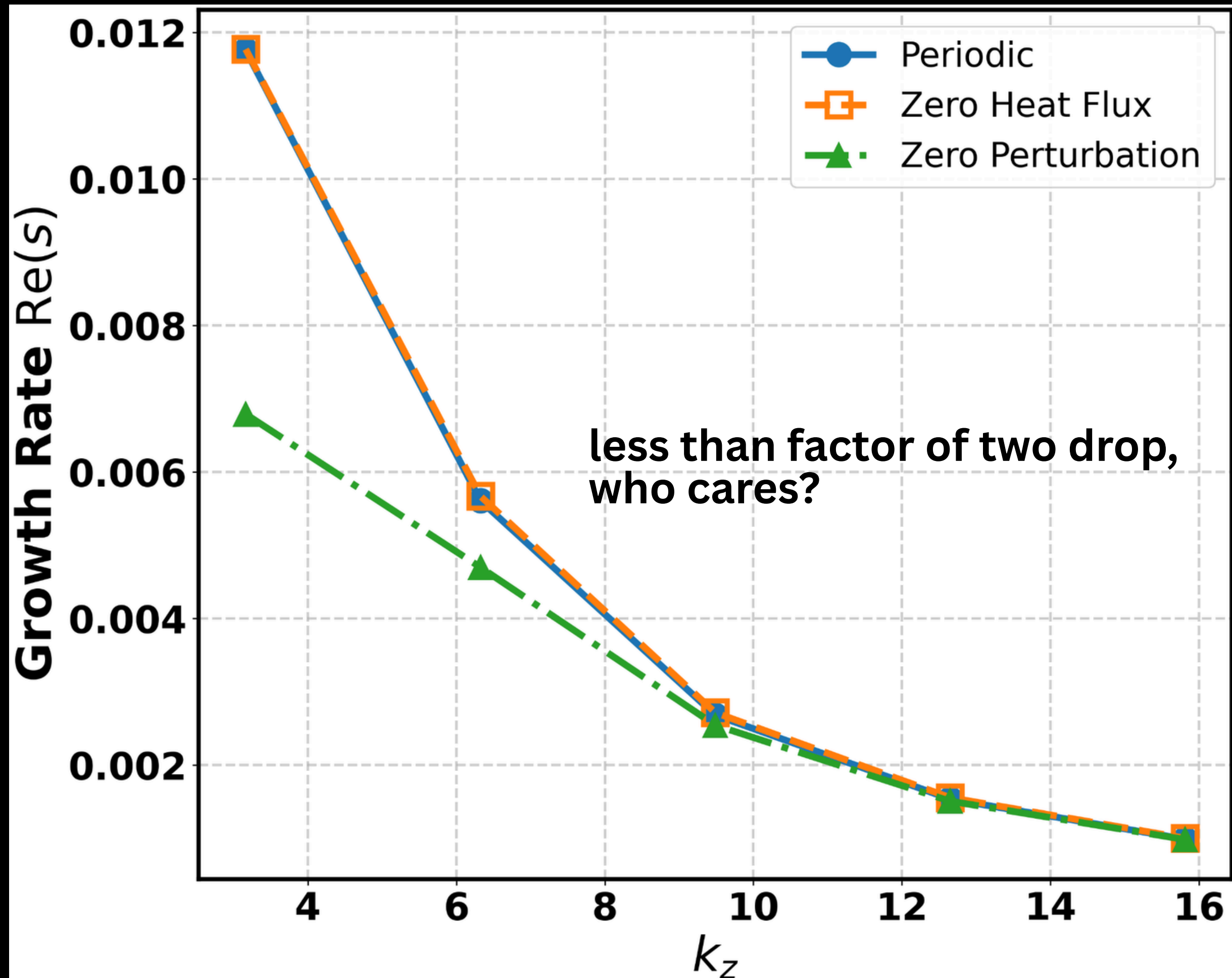
Standard periodic boundaries



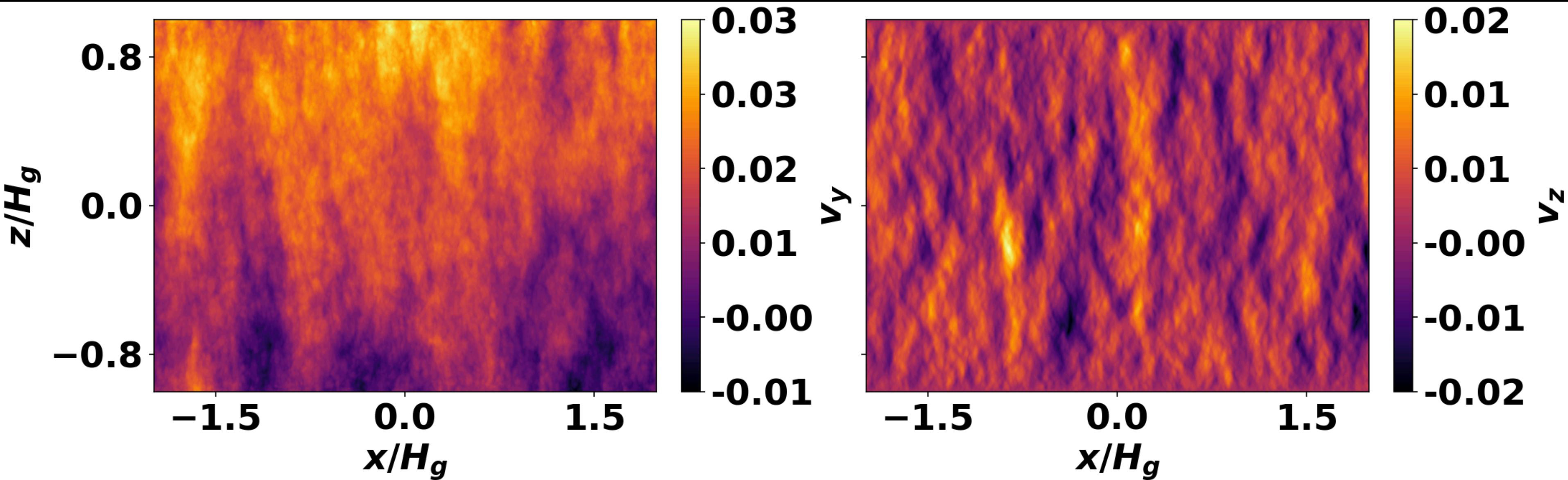
Zonal & elevator flows

Periodicity isn't a must

- solve vertically global eigenvalue ODE



Zero perturbation in z

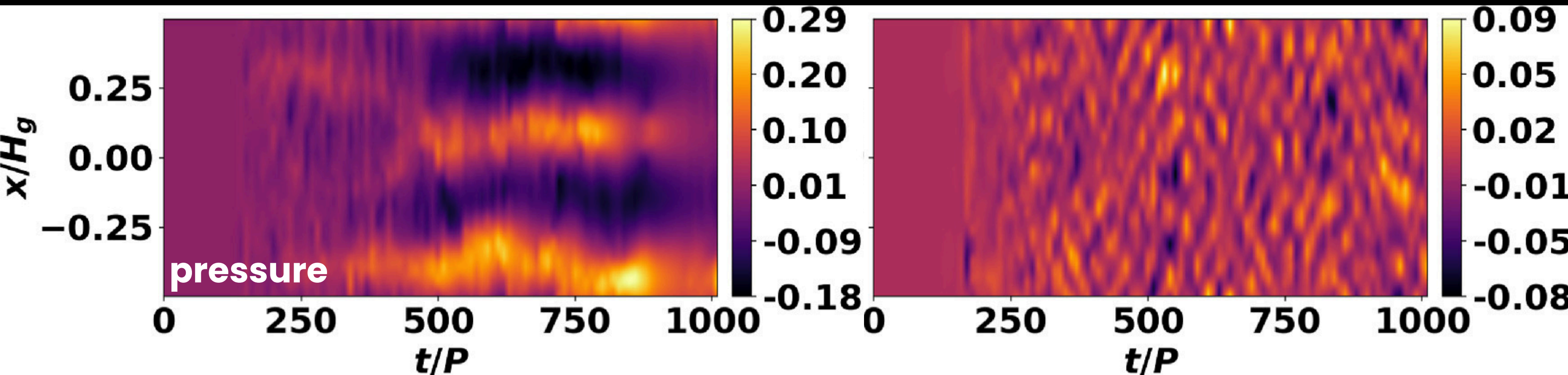


No structures

Dust can also surpress zonal flows

$$\langle \rho_d / \rho_g \rangle = 0.01$$

$$\langle \rho_d / \rho_g \rangle = 0.1$$



Need negative ang. mom. flux ($\alpha < 0$) for ZF, but $\alpha_{\text{dust}} > 0$
(Teed & Latter, 2021)

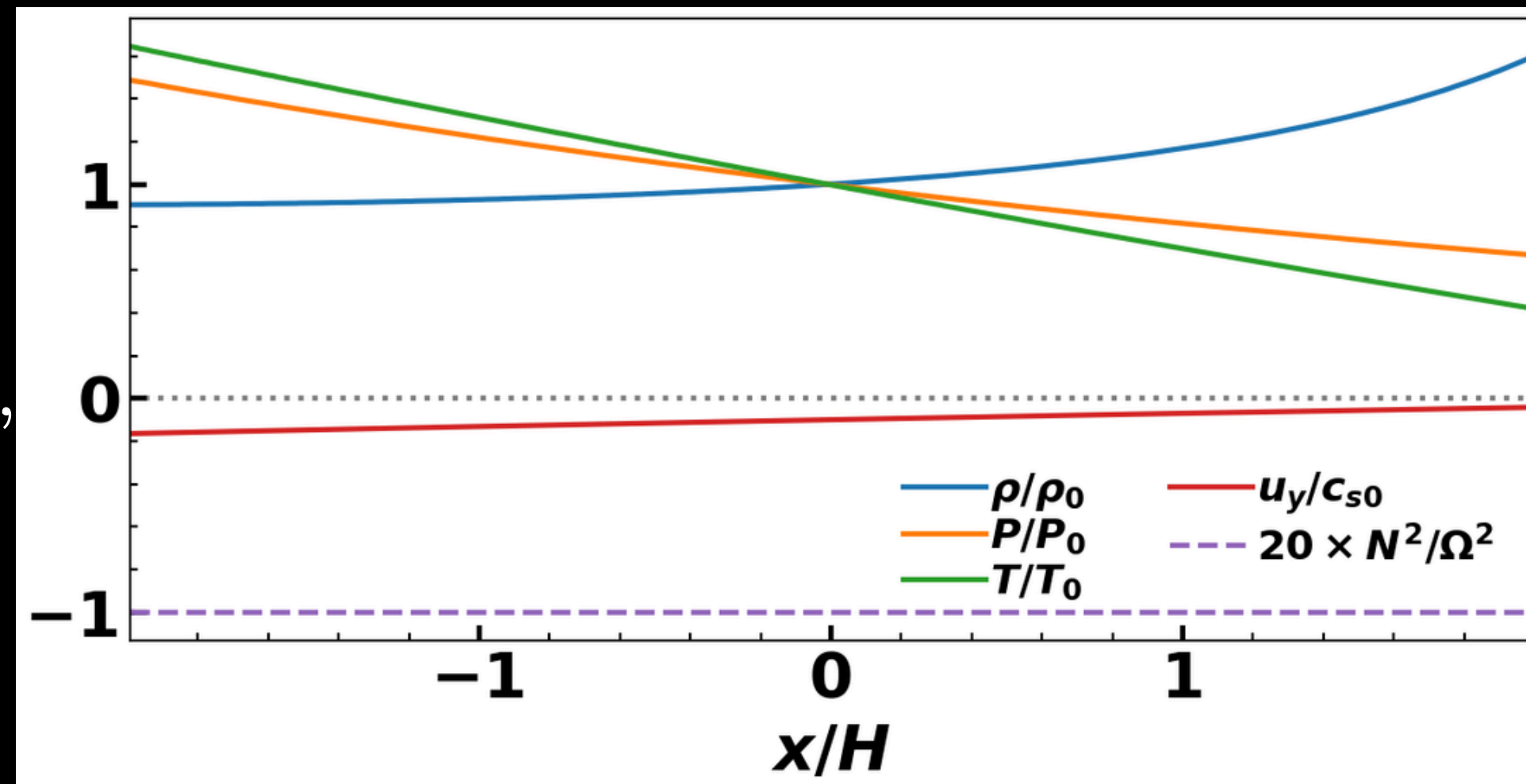
Locally global model

i.e. radially stratified box

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0,$$

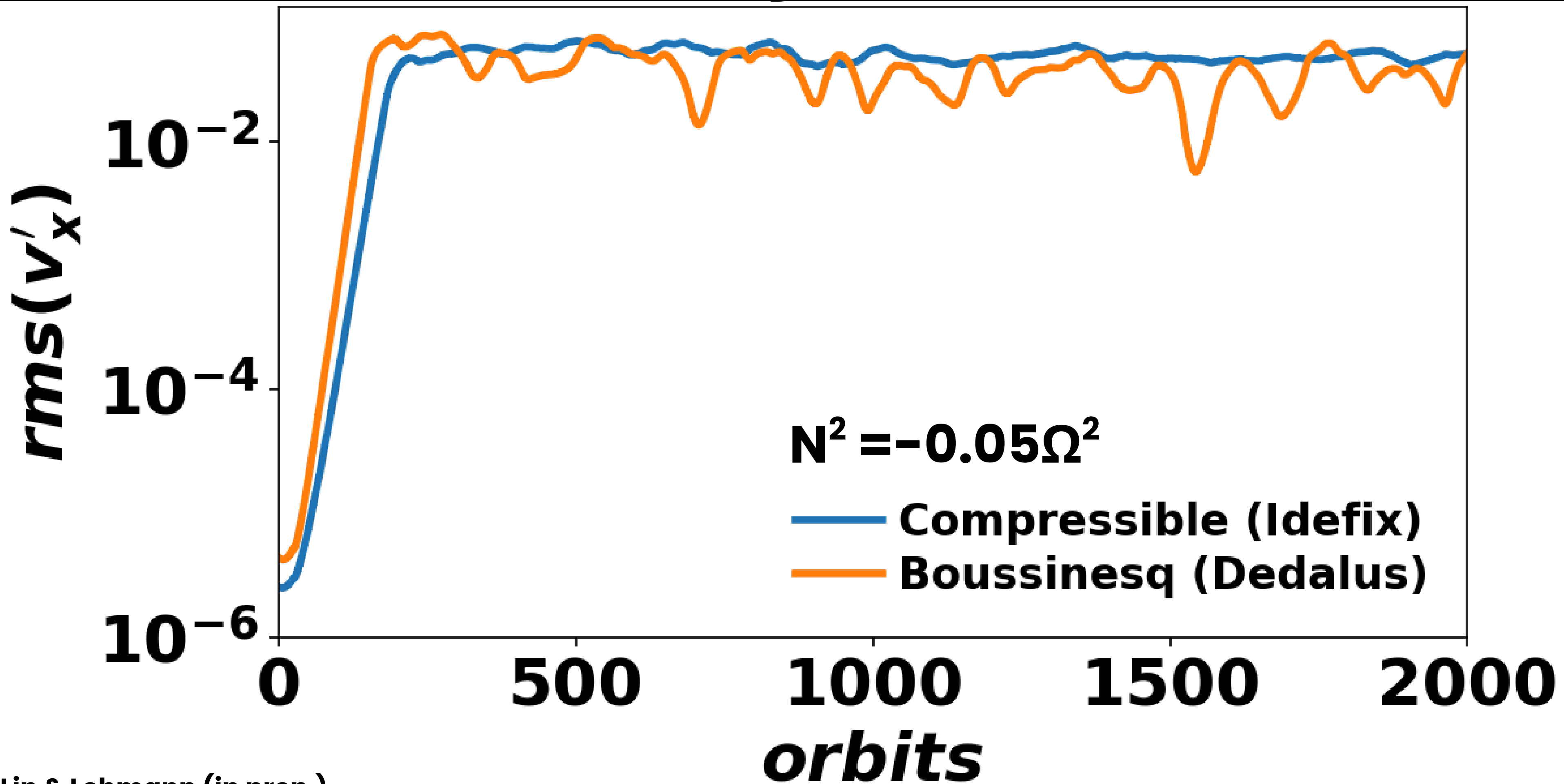
$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = 3\Omega^2 x \hat{\mathbf{x}} - 2\Omega \hat{\mathbf{z}} \times \mathbf{v} - \frac{\nabla P}{\rho} + \frac{\nabla \cdot \mathcal{T}}{\rho},$$

$$\frac{\partial e}{\partial t} + \nabla \cdot (e \mathbf{v}) = -P \nabla \cdot \mathbf{v} + \kappa_T \nabla^2 T' + Q'_{\text{visc}},$$

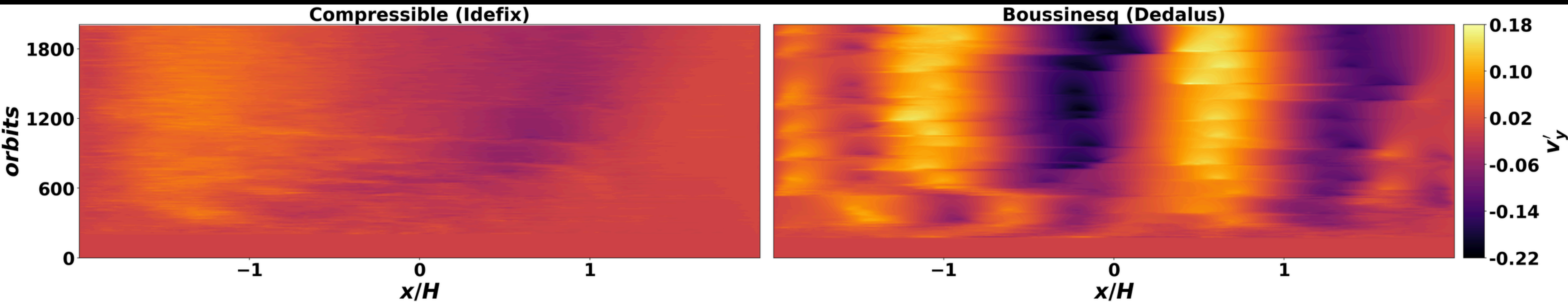


- Specify $N^2(x) \rightarrow \rho(x)$, assuming smooth $P(x)$
- zero heat flux radial boundaries, vertically periodic
- Fix $\alpha_{\text{visc}}=10^{-5}$
- Idefix (Lesur et al 2023), tested against linear theory

Growth and saturation



Compressibility weakens zonal flows

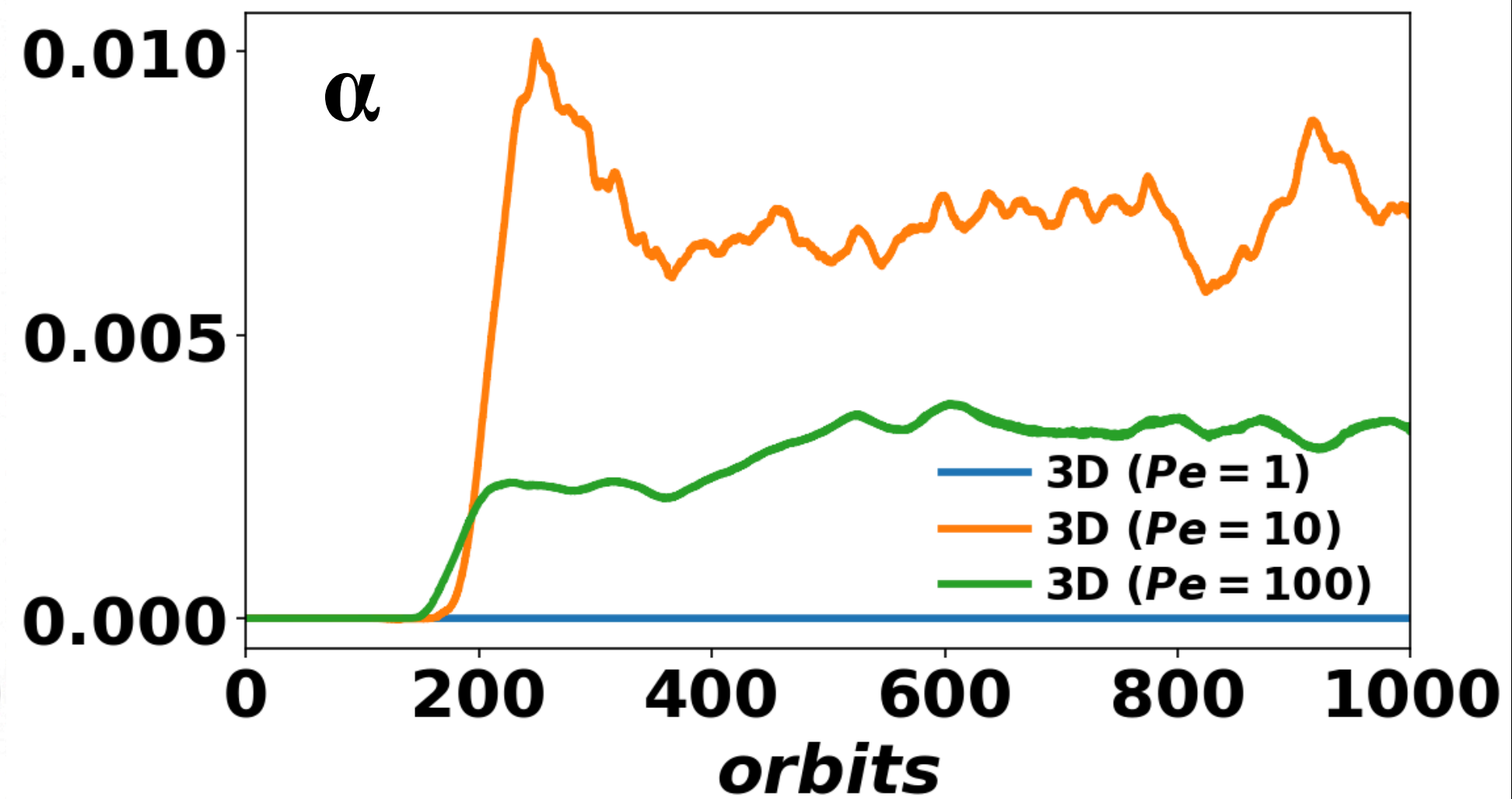
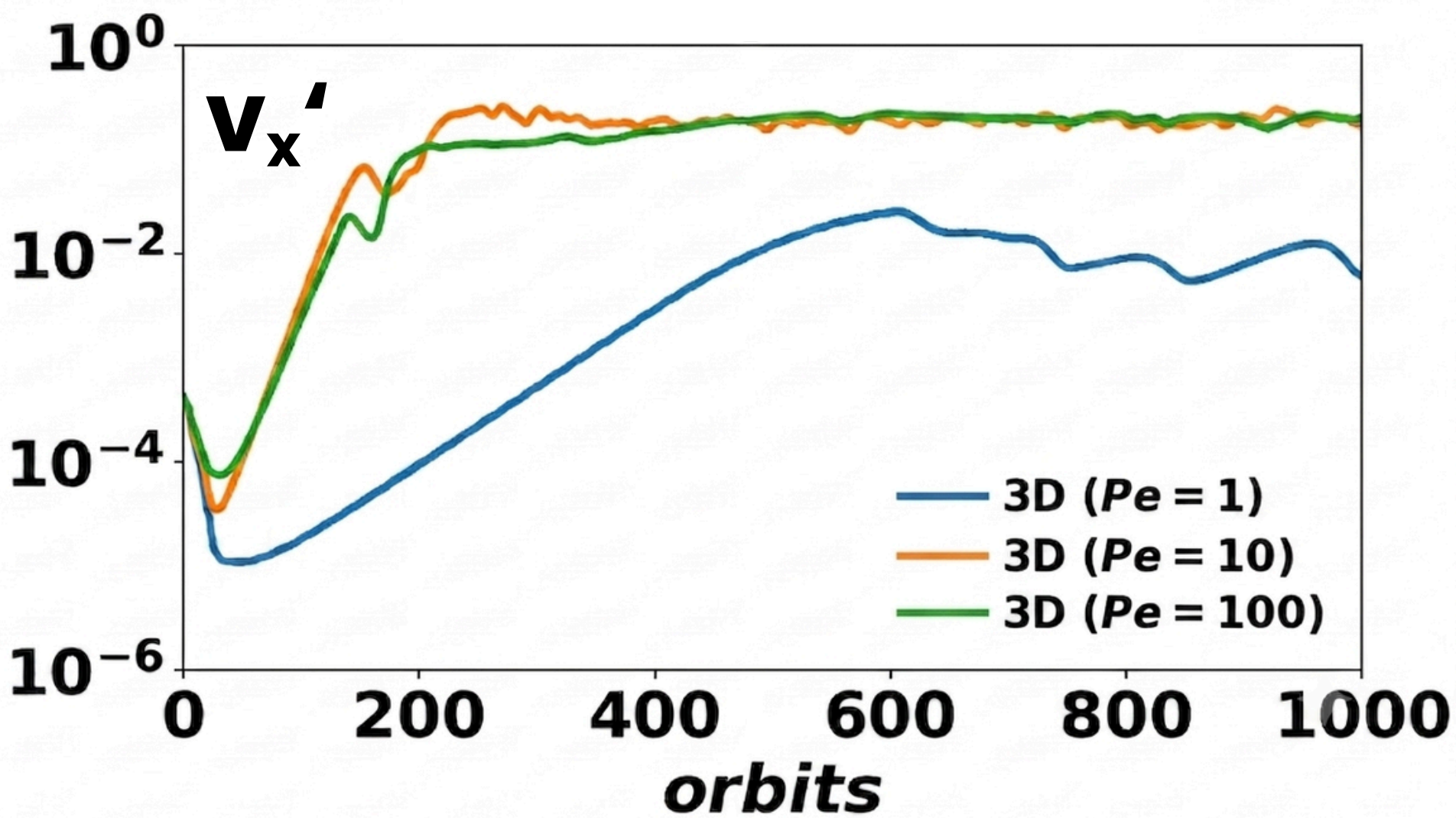


weak zonal flows

clear zonal flows

3D, variable thermal diffusion

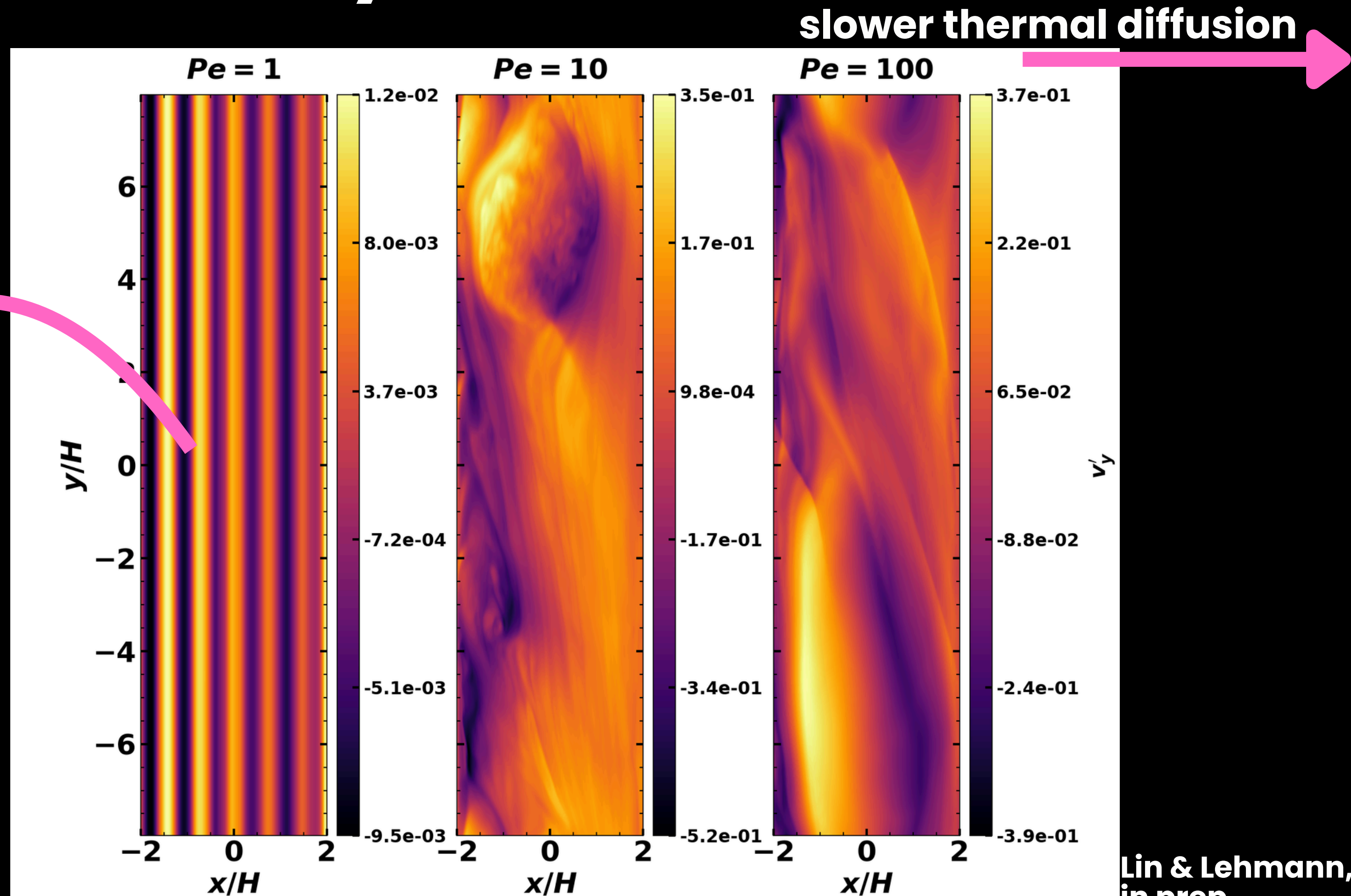
Peclet number = $H^2 \Omega / \xi$, larger $Pe \rightarrow$ slower diffusion



Vortices don't always form

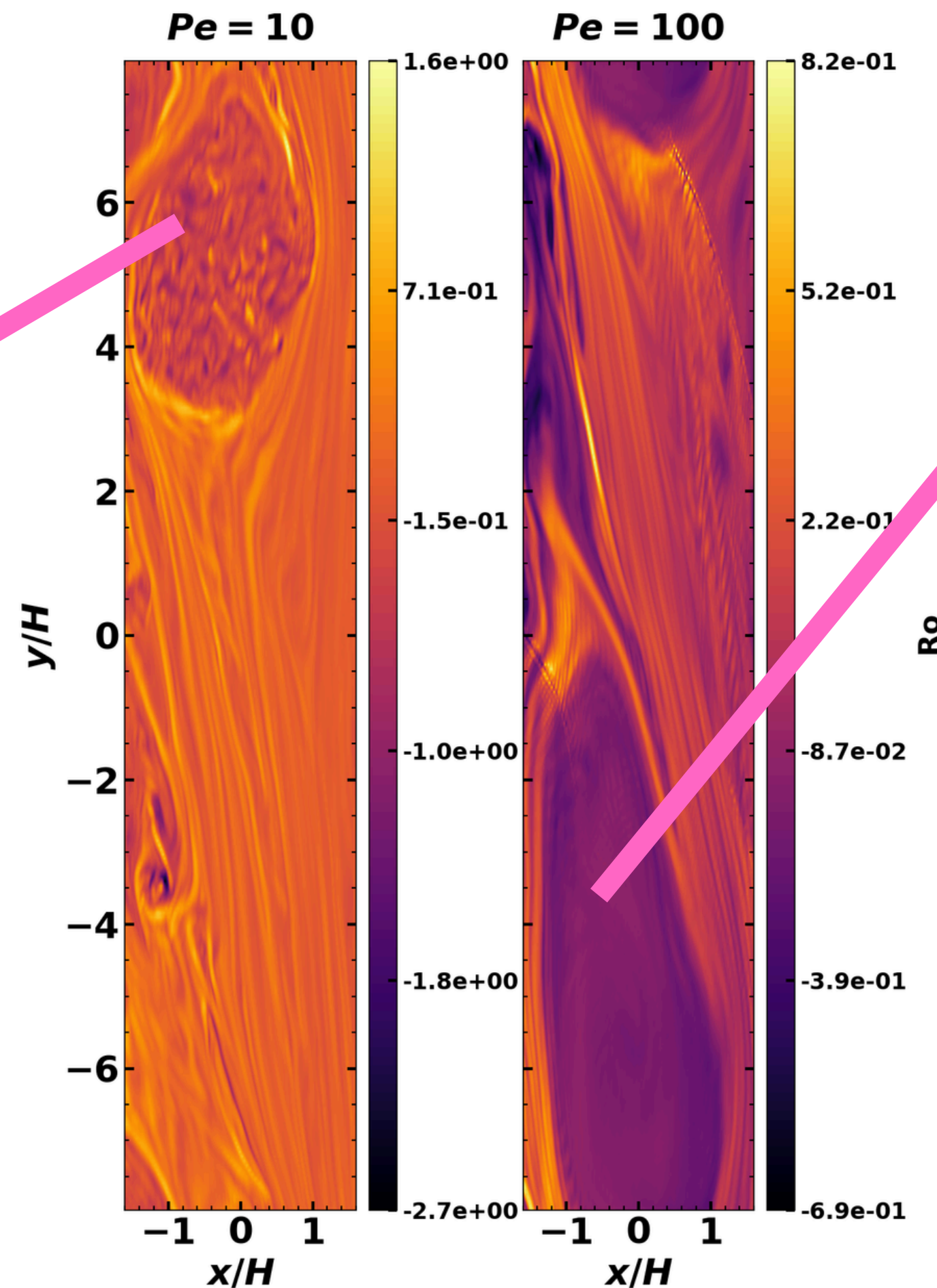
weakly unstable
disks stay
axisymmetric

cf. Teed & Latter (2025)



Which vortex is more useful?

**compact, strong
vortex with internal
turbulence**
(Lesur & Papaloizou, 2009)

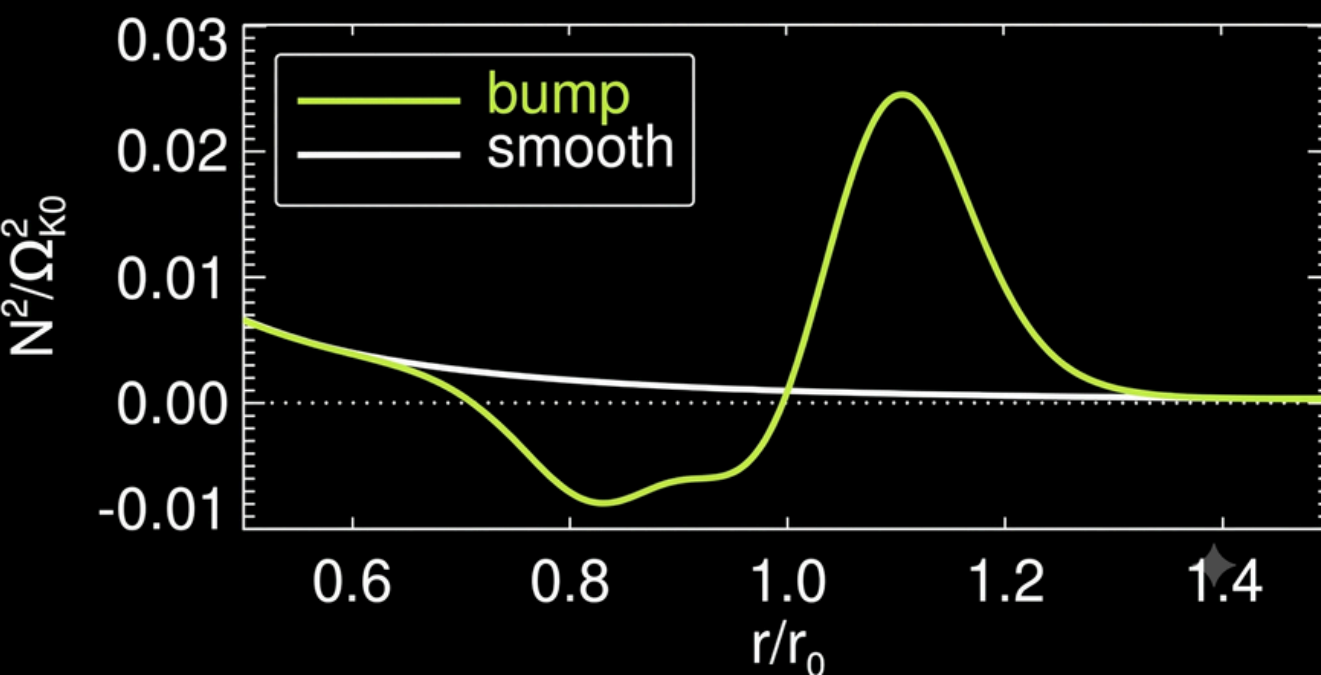


**elongated, weak
vortex, but internally
laminar**

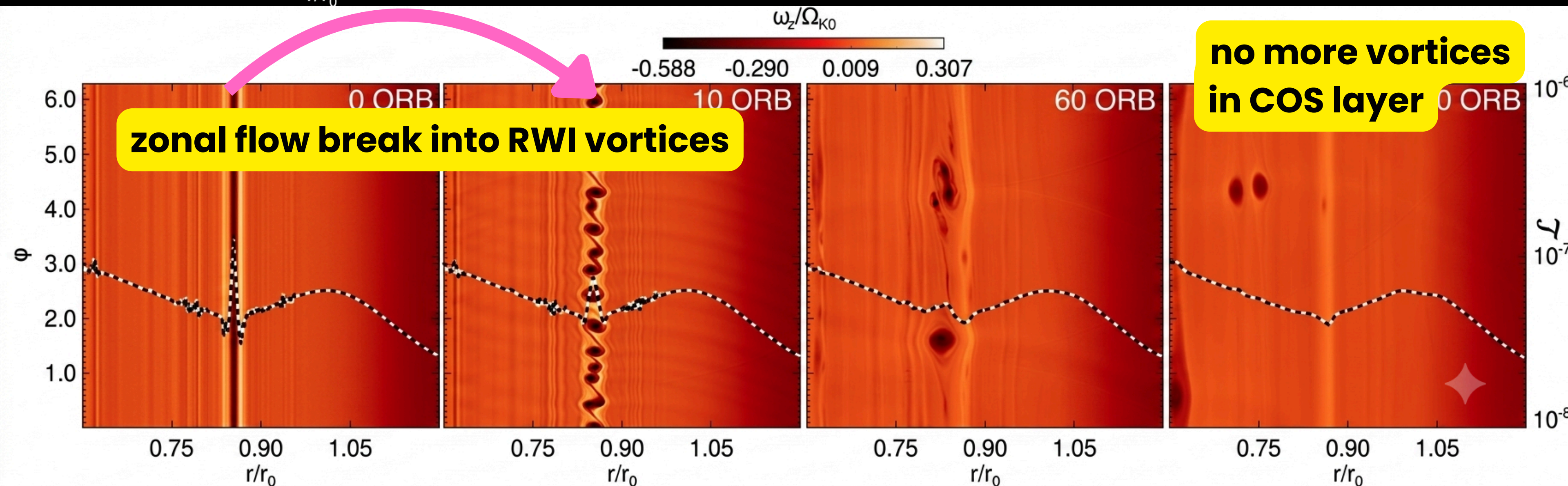
$N^2 < 0$ may not apply across large radii in midplane

What if $N^2 < 0$ is localized?

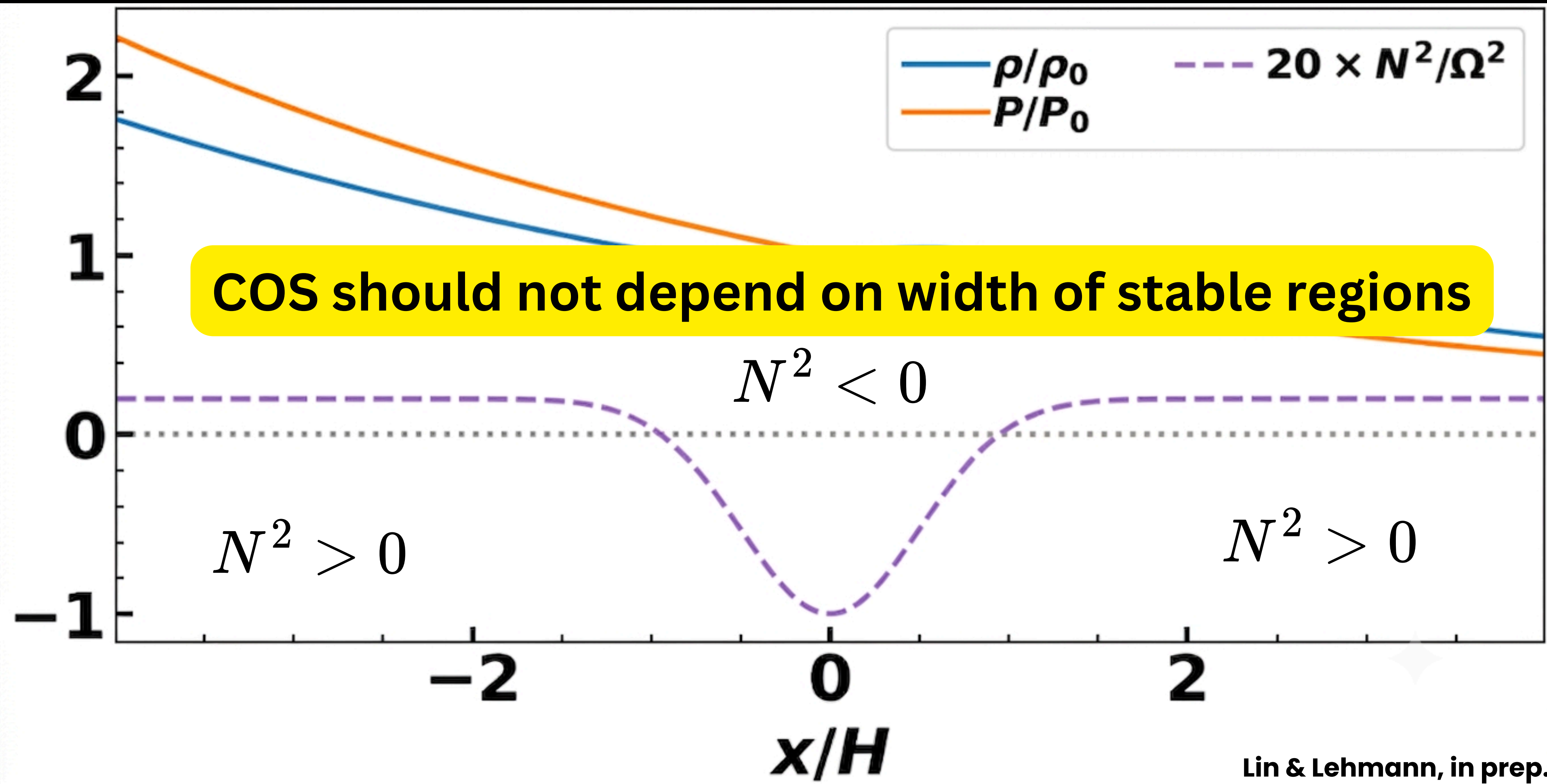
A cylindrical first attempt



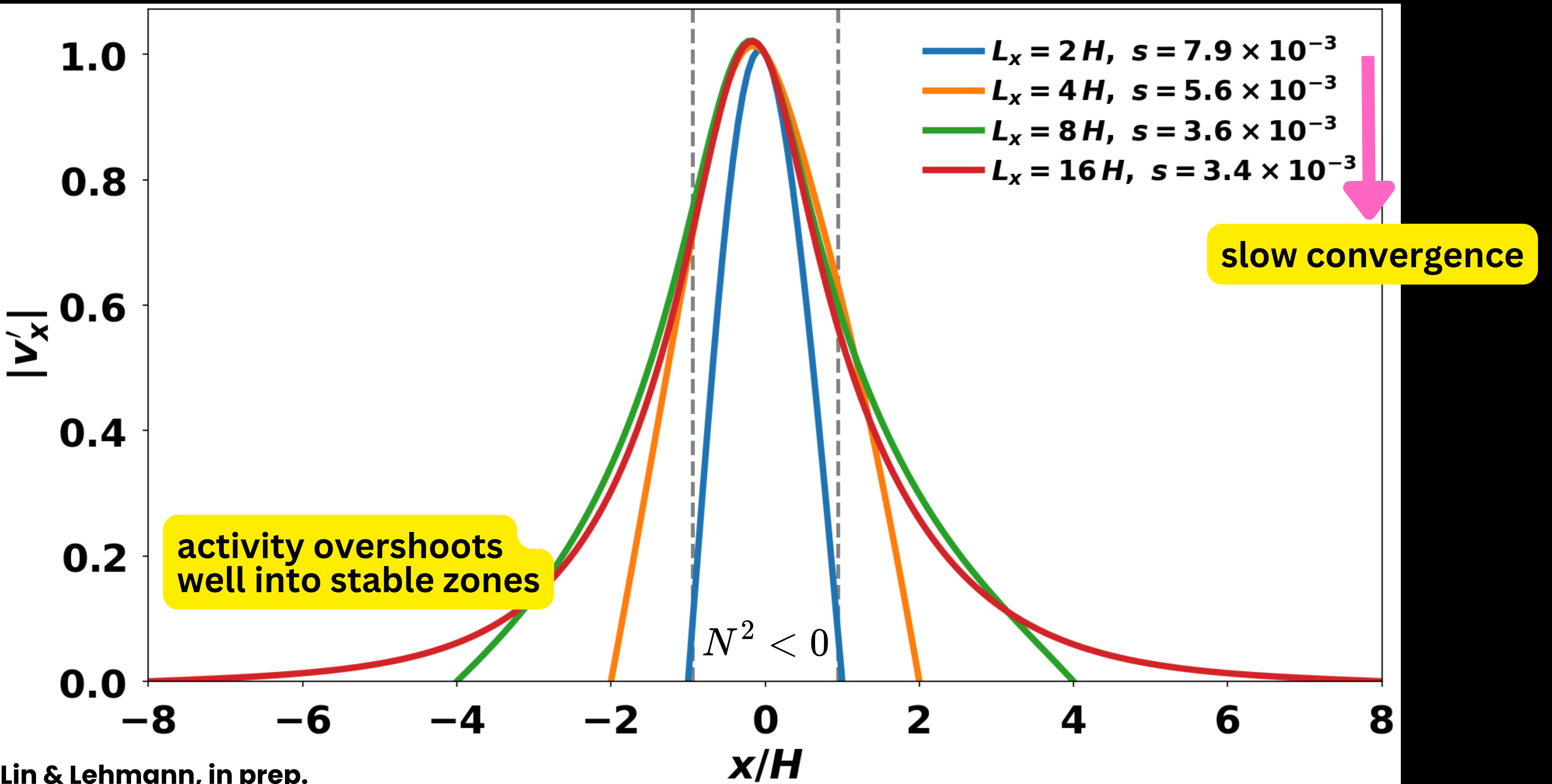
- 2D for 30,000 orbits, restart in 3D



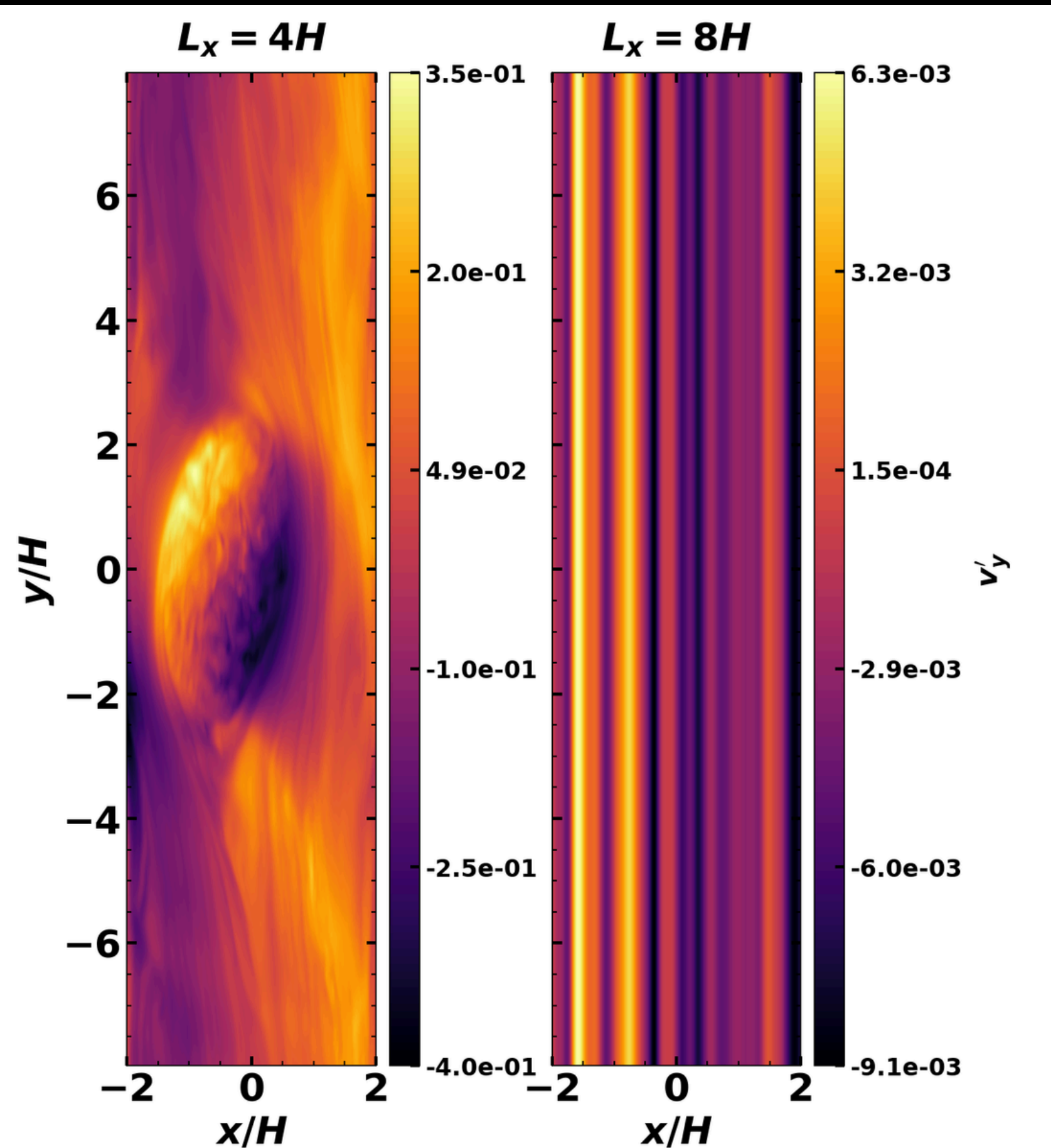
Locally global setup



Global modes for local features

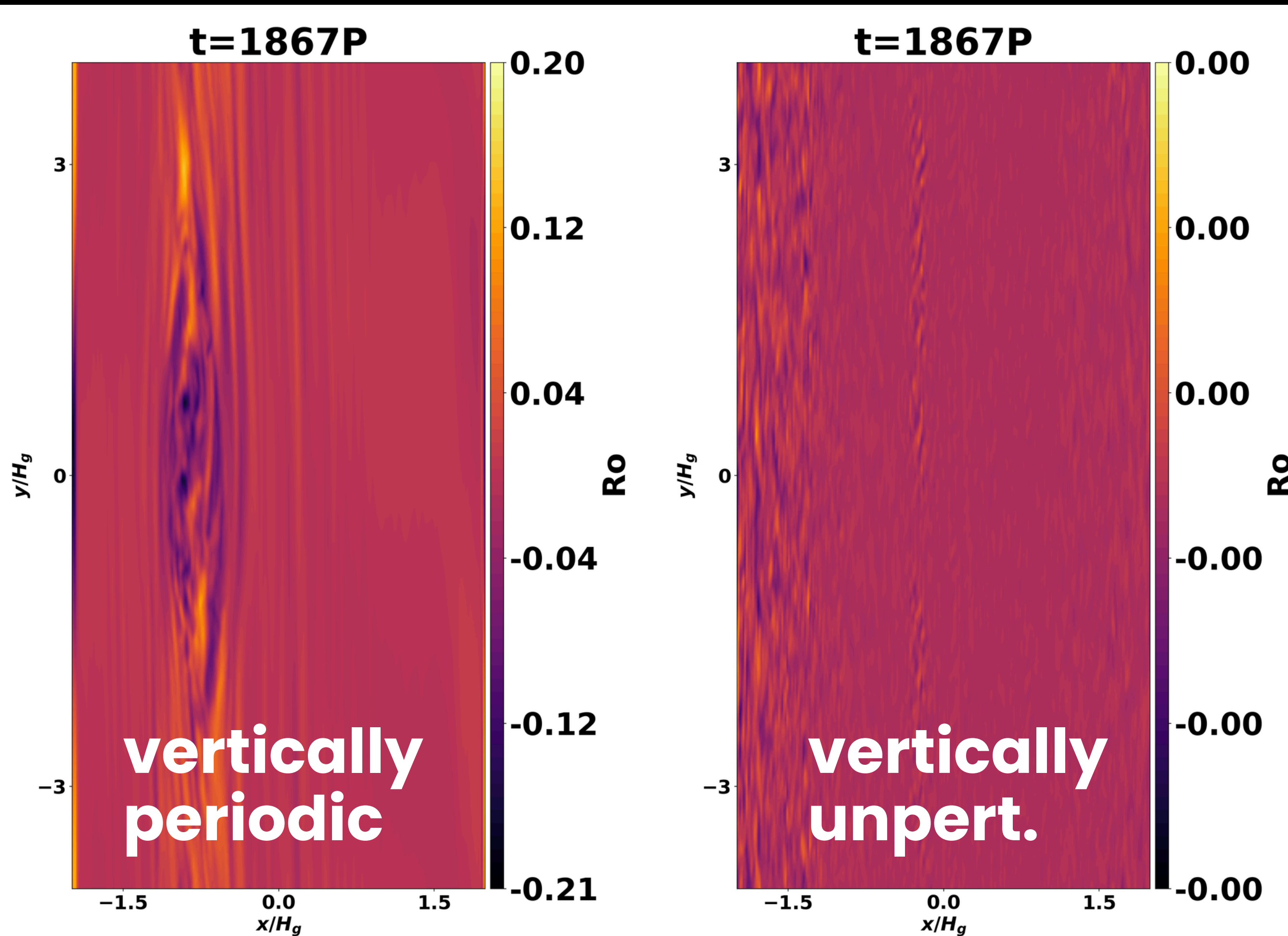


Small boxes or const. N^2 might mislead



- Identical N^2 profiles
- Narrow box $\rightarrow$ vortex
- Wide box $\rightarrow$ rings

Impact of vertical boundaries



Summary

- **Non-shearing boxes with self-consistent disk evolution**
- **Precise control over initial $N^2(x)$**
- **Local N^2 features still need extended domains**
- **COS need not produce vortices; even then, boundaries can have an impact**

Thank you
X @minkailin